Classical fluctuation laws and conditional action universality

Independent centre-of-mass composition makes a **nonnegative, mass-only** fluctuation coefficient the same $K \geq 0$ for every positive mass, provided a composite and a whole-body preparation of the same mass have the same assigned coefficient. This premise does not select positivity. A supplied finite-rate, nonzero-speed reference gives $K > 0$ within its preparation class; matched mechanical parameters without matched correlations can instead give zero.

Continuous free paths with stationary independent increments and isotropic centred noise form a Brownian subclass, with variance rate $D(m) = K/m$. For positive K, additional additive force-independent position noise gives a sharp pinned-path record floor. Neither this stochastic law nor composition alone gives quantum dynamics, universal preparation equivalence, or the identification $K = \hbar$. Physical $\hbar$ is a separate calibrated input. All proofs and source-reading qualifications below are internal written work, not independent peer review or evidence of literature novelty.

Observable and independent composition

All masses are positive. Work on the line for the covariance formulas, or coordinatewise for isotropic noise. For centred independent displacements over a common window $\Delta > 0$, define $\mathrm{Var}(\Delta X_i) = \kappa_i \Delta / m_i$, where κ_i has action units. Put $M = m_1 + m_2$, $\mu = m_1 m_2 / M$, $R = (m_1 X_1 + m_2 X_2)/M$ and $r = X_1 - X_2$. Variance addition gives

$$\kappa_{\mathrm{cm}} = \frac{M}{\Delta}\,\mathrm{Var}(\Delta R) = \frac{m_1\kappa_1 + m_2\kappa_2}{M}, \qquad \kappa_{\mathrm{rel}} = \frac{\mu}{\Delta}\,\mathrm{Var}(\Delta r) = \frac{m_2\kappa_1 + m_1\kappa_2}{M}.$$

Consequently $\kappa_{\mathrm{cm}} + \kappa_{\mathrm{rel}} = \kappa_1 + \kappa_2$ and $\mathrm{Cov}(\Delta R, \Delta r) = \Delta(\kappa_1 - \kappa_2)/M$. For independent Gaussian bridges, replace Δ by their common covariance kernel. Centre and relative processes are independent exactly when $\kappa_1 = \kappa_2$ in that Gaussian setting. More generally, for $y = \sum_i c_i X_i$, use the kinetic mass $\mu_y = (\sum_i c_i^2/m_i)^{-1}$; its coefficient is the weighted average

$$\kappa_y = \frac{\sum_i (c_i^2/m_i)\kappa_i}{\sum_i c_i^2/m_i}.$$

This transformation is not a mechanism selecting a positive value.

A distinct observable comes from stationary finite irreducible reversible chains J^i with centred velocities $v_i(J^i)$ and $X_i(t) = \int_0^t v_i(J_s^i)ds$. C019 defines the Green–Kubo plateau $H_i = 2m_i \int_0^\infty C_i(s)ds$. Independence gives $C_{\mathrm{cm}} = (m_1^2 C_1 + m_2^2 C_2)/M^2$, hence the same coefficient formulas for H. The product generator is $Q_1 \otimes I + I \otimes Q_2$, with invariant law $\pi_1 \otimes \pi_2$; irreducibility and detailed balance hold factorwise. Keep the full product state when distinct state pairs share a velocity: its velocity projection need not be Markov. If each speed is at most $u < c$, the centre speed is too. Relative speed can reach $2u$, since it is a coordinate difference, not an additional particle subject to the same speed premise.

Mass-universality theorem

Assumptions. Within one preparation class, every real mass $m > 0$ has a finite coefficient $\kappa(m) \geq 0$ depending only on mass. Independent constituents may be composed, and their assigned whole-body coefficient equals the centre coefficient above. Use the same observable throughout: a common-window variance, a Gaussian covariance coefficient, or a Green–Kubo plateau. The argument does not exchange these windows or limits.

Conclusion (S2). There is one constant $K \geq 0$ with $\kappa(m) = K$ for all $m > 0$.

Proof. Let $f(m) = m\kappa(m)$. Composition says $f(a + b) = f(a) + f(b)$ on the positive reals. Since $f \geq 0$, $f(b) - f(a) = f(b - a) \geq 0$ for $b > a$. Fix a mass unit $m_0 > 0$. Additivity gives $f(qm_0) = qf(m_0)$ for positive rational q. Rational sequences approaching m/m_0 from below

and above sandwich $f(m)$ by monotonicity, giving $f(m) = mf(m_0)/m_0$. Thus $K = f(m_0)/m_0$, including $K = 0$. No additional measurability or empirical mass-continuity premise is needed on this domain. □

The all-positive-mass domain and equivalence of independent composite and whole-body preparations do the physical work. A restricted species list needs its own domain theorem. An environmental label θ permits a different $K(\theta)$ at each fixed label. Correlated displacements add $2m_1m_2\,\mathrm{Cov}(\Delta X_1, \Delta X_2)/(M\Delta)$ to the centre coefficient; bound bodies and common reservoirs require that actual correlation calculation. The interaction/rotation theorem concerns canonical commutators and a common Newton derivation, rather than this probabilistic preparation premise; it independently retains a zero branch.

Continuous free paths and ultimate velocity

Assumptions (S1). A body's free motion starts at a deterministic position, has almost surely continuous paths and stationary independent increments, and has a rotation-invariant centred fluctuation law in $\mathbb{R}^3$. A constant drift is allowed; isotropy of the full law in a fixed frame would instead force that drift to vanish. Universality of free fall alone gives none of these noise premises.

Continuity supplies stochastic continuity. The continuous Lévy classification removes the jump part, leaving

$$X_t = x_0 + vt + \sqrt{D(m)}W_t, \qquad D(m) \geq 0.$$

Isotropy makes the covariance matrix $D(m)I$. If the mass-only and independent composition premises also hold, the coefficient in the preceding theorem is $\kappa(m) = mD(m)$, so $D(m) = K/m$. Equivalently, $g(m) = m^2D(m)$ is nonnegative additive; this is the same proof, not another selection principle.

For $D = 0$ the paths are inertial. For $D > 0$ Brownian paths are almost surely nowhere differentiable, while $\lim_{h\downarrow 0}\mathbb{E}[X_{t+h} - X_t \mid \mathcal{F}_t]/h = v$. Their spatial range over a nontrivial interval in three dimensions has Hausdorff dimension two. A sharp position at an instant and no pathwise instantaneous velocity are therefore compatible in this model; this is not the quantum state uncertainty theorem.

Newton's scholium to Book I, Section I calls the ultimate velocity "certus & definitus"; the primary-text discussion gives the 1687 passage and its reading. Positive Brownian noise changes that pathwise premise while retaining a mean forward velocity. This is a stated modern comparison, not evidence that Newton anticipated Brownian or quantum kinematics. The proposed continuity comparison with Aristotle (*Physics* VI, book-level citation) is likewise historical interpretation; jump processes below are a mathematical alternative, not a claim of transmission from ancient "leap" accounts.

Positive reference theorem

For Green–Kubo plateaus, suppose the class includes one stationary symmetric two-state reference with mass m_0, velocities $\pm u_0$, $0 < u_0 < c$ and reversal rate $0 < \lambda_0 < \infty$. C020 gives

$$\boxed{K = H(m_0) = \frac{m_0u_0^2}{\lambda_0} > 0, \qquad H(m) = K \quad (m > 0).}$$

The value and positivity come from the reference's nonzero speed and finite clock, together with the composition premise. Other two-state members obey $\lambda_m = mu_m^2/K$. For any chosen $K > 0$ and common $0 < u < c$, independent members with $\lambda_m = mu^2/K$ and all finite products realize the same mass-normalized plateau. Their centre velocities need not have two values. This is a consistency example, not a derivation of the assigned clock law. Across such classes

K approaches zero while every individual rate remains finite. A numerical lower bound needs uniform reference controls, for example $u_0 \geq u_{\min} > 0$ and $\lambda_0 \leq \Lambda < \infty$ at fixed m_0.

Bounded-speed microscopic paths have vanishing short-window variance coefficient (C018), even when their long-window plateau is positive. The return bridge connects windows within one path law; a Brownian limit exchanges a microscopic clock and observation scale. Checkerboard dynamics adds a distinct coherent propagator premise. No such continuation follows from covariance alone. The finite-chain spectral and limit proofs remain in classical-action-field, PDF.

Preparation and correlation tests

In A02, independently prepared tracers in otherwise identical refreshed baths have $H(m) = (m + M_b)s^2/\nu$. Treating their total mass as a single tracer gives

$$H(m_a + m_b) - \frac{m_a H(m_a) + m_b H(m_b)}{m_a + m_b} = \frac{2m_a m_b s^2}{\nu(m_a + m_b)}.$$

For $s > 0$ this positive discrepancy refutes preparation equivalence for those experiments, while preserving the bath calculation. An actual bound composite needs its own mechanical model.

C050–C051 give a stronger matched-scale test. In the equal-mass hard-point gas, fix $m, \rho, u > 0$. Independently marked Poisson streams yield reversal rate ρu and plateau $H_* = mu/\rho$. Two ordered counterpropagating streams with the same mass, speed, density, mean collision frequency and energy density instead give a Palm-tagged uniformly phased square-wave velocity, with period $2d$, $d = 1/(\rho u)$. Displacement cancels each period and

$$0 \leq \frac{m}{\Delta} \operatorname{Var}[X(t + \Delta) - X(t)] \leq \frac{m}{3\rho^2 \Delta} \longrightarrow 0.$$

The preparations differ in joint position–velocity correlations and mark independence; the ordered process is not mixing. The collision manuscript, §§5–6, PDF, contains both proofs and the Palm residual-time calculation; B26 and its derivation record the comparison. Matching these mechanical scales establishes no universal floor.

A conditional Brownian record floor

For S3 add an explicit force law: under constant force F, the path is the Newtonian parabola plus the **same additive force-independent position noise** $\sqrt{K/m}W_t$. This extra premise is not part of the free classification. Let $m, \tau, K > 0$, take equal priors, and condition both forced and inertial laws on the same cell endpoints. Their mean difference is $\delta x(u) = Fu(\tau - u)/(2m)$ and their bridge covariance is $(K/m)(\min(u, v) - uv/\tau)$. With $\mathcal{K}_\tau = F^2\tau^3/(24m)$, the Cameron–Martin norm and optimal error are

$$\|\delta x\|^2_{\mathrm{CM}} = \frac{m}{K} \int_0^\tau \dot{\delta x}^2 du = \frac{2\mathcal{K}_\tau}{K}, \qquad P^{\mathrm{opt}}_{\mathrm{err}} = \Phi\left(-\sqrt{\frac{\mathcal{K}_\tau}{2K}}\right).$$

The cut-measure proof gives the Gaussian likelihood calculation and cut evidence decomposition: a cut at fraction s of a cell of length L carries Kullback–Leibler share $3s(1-s)\mathcal{K}_L/K$. Any common observation channel applied to the path can only increase Bayes error. For $0 < \varepsilon < 1/2$, confidence $1 - \varepsilon$ requires $\mathcal{K}_\tau \geq 2z^2_{1-\varepsilon}K$. At $F = 0$ the error is $1/2$. For $F \neq 0$, **if** $K = \hbar$ is supplied, this becomes

$$\tau \geq \left(\frac{48 z^2_{1-\varepsilon} m\hbar}{F^2}\right)^{1/3},$$

the Planck working manuscript's pinned-path mesh. Here $\hbar$ is the coefficient in $e^{-S/\hbar}$, not a mesh parameter. This classical stochastic model with an assigned noise coefficient does not explain why that coefficient should equal physical $\hbar$.

Jumps allowed: the composition law beyond Brownian motion

Replace path continuity by stochastic continuity with a càdlàg version, retain stationary independent increments and deterministic initial point, and drop the finite-variance requirement. Assume symmetric isotropic centred Lévy laws, with real nonnegative exponent

$$\mathbb{E}e^{ik\cdot(X_t-X_0)} = e^{-t\phi(m,|k|)}.$$

Composition requires $\phi(M,|k|) = \sum_i \phi(m_i, m_i|k|/M)$. For fixed $p = |k|/M$, put $G(m,p) = \phi(m,mp)$. It is nonnegative additive in m, so the same rational sandwich argument yields, with one universal Lévy exponent Γ,

$$\phi(m,|k|) = m\Gamma(|k|/m), \qquad \nu_m(A) = m\nu(mA).$$

Here a reference mass unit is understood in writing $\Gamma(p) = G(1,p)$; ν is its Lévy measure. Jumps shrink by $1/m$ while rates multiply by m. Brownian motion has $\Gamma(p) = Kp^2/2$. Symmetric α-stable examples have $\Gamma(p) = K_\alpha p^\alpha$ and coefficient $K_\alpha m^{1-\alpha}$. This is S4's composition classification, not a positive-noise theorem.

A separate dimensional statement assumes there are **no other dimensional inputs or hidden scales**, and assumes dimensional covariance of the law. Since p has units $(\text{mass}\,\text{length})^{-1}$ and Γ has units $(\text{mass}\,\text{time})^{-1}$, a sole action constant K permits only $\Gamma(p) = CKp^2$ with dimensionless $C \geq 0$. Under those additional premises the law is Brownian. Composition alone does not impose the single-scale assumption.

An additional speed c allows $\Gamma(p) = Kp^2 f(Kp/c)$. For $K > 0$, choose

$$\Gamma(p) = \frac{\sqrt{c^4 + c^2K^2p^2} - c^2}{K}, \qquad \phi(m,|k|) = \frac{\sqrt{m^2c^4 + c^2K^2k^2} - mc^2}{K}.$$

This exponent is a Bernstein function of k^2 and generates jumps; its small-$K|k|/(mc)$ limit is $Kk^2/(2m)$. It is the relativistic kinetic energy divided by K in a **Euclidean generator**. Instantaneous sample-path jumps remain allowed, so c supplies no causal speed bound on these paths.

Mean dynamics, historical interpretation and source evidence

Nelson's coefficient convention is $\nu = K/(2m)$. His stochastic mechanics imposes a Newton law on a symmetric mean acceleration, using forward/backward mean derivatives and state-dependent drift. The free assumptions here give neither that dynamics nor a wave function. Phase/circulation conditions are additional obligations. In the additive-force model one has $m\,d^2\mathbb{E}X_t/dt^2 = F$ before endpoint conditioning; that elementary identity differs from Nelson's symmetric mean law. Literal pathwise Newtonian momentum requires reformulation for Brownian paths. Unqualified equivalence to the Newtonian Laws, or derivation of Schrödinger dynamics from the free and composition premises alone, is withdrawn.

- Lalley, *Introduction to Brownian Motion*, §1 was read at passage level for the continuous-Lévy classification. Paley–Wiener–Zygmund (1933) and Abbott–Wise (1981) are metadata-level leads for nondifferentiability and the path-dimension comparison respectively.

- Nelson (1966), abstract, supplies $\nu = \hbar/(2m)$ and reports a limited stochastic-mechanics equivalence; Fényes (1952) is a precursor identified by metadata. Wallstrom (1994), abstract, records the phase/global-equivalence issue. These reading levels support no full-proof audit of stochastic mechanics.
- Demme–Caticha (2017), §3, equations (18)–(28), computes the centre covariance from supplied inverse-mass constituent covariances in its equation (9). That passage is close prior art for composition, rather than proof that the converse or a positive universal constant is new. The bounded original search located no explicit converse in the passages checked and establishes no priority claim.
- Pavliotis (2010) supplies the Poisson-equation input behind C019–C020; Pitman–Yor (2018) supplies Gaussian covariance background. Per-result prior-art qualifications remain in B16 and the independent manuscripts.
- Carmona–Masters–Simon (1990) is a metadata-level lead for the relativistic-generator comparison.

Consequence for STATE

The common-coefficient theorem, continuous Brownian subclass, finite-clock reference and jump classification now share one composition argument. Positivity, whole-body preparation equivalence, the actual force/noise law, quantum reconstruction and calibration to physical $\hbar$ remain different obligations. The ordered-stream and finite-window countermodels must accompany any use of a positive plateau. New research remains suspended during consolidation.